

Anshu Yadav

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PROFESSIONAL SUMMARY

I am a dedicated applied researcher specializing in the projection and utilization of wave power energy, with a focus on the Indian and Indo-Pacific Ocean. My expertise lies in analyzing the impact of oceanic warming, driven by increasing sea surface temperatures, on wave power energy. I utilize advanced climate models, such as CMIP5, CMIP6 and the Coordinated Ocean Wave Climate Project (COWCLIP), to evaluate the models and further project wave power and examine the influence of climate phenomena like ENSO and IOD on wave energy. My research aims to identify key locations along the Indian coastline for renewable wave power generation, contributing to sustainable energy solutions with minimal greenhouse gas emissions. I have experience working with observational data, satellite information, and buoy measurements, and I am passionate about providing high-quality research to advance the field of ocean energy.

RESEARCH INTERESTS

EDUCATIONAL QUALIFICATIONS

Course/Examination	Institution / University	Year of Passing	Percent
Ph. D. in Applied Sciences (Mathematics)	NIT Delhi, New Delhi	Ongoing	9.5 CGPA
Masters in Mathematics	NIT, Jalandhar	2021	8.04 CGPA
Bachelor of Science (Mathematics Honors)	Gargi College, University of Delhi	2019	7.64 CGPA
Class XII	CBSE Board	2016	90.8 %
Class X	HBSE Board	2014	95.6 %

ACHIEVEMENTS & AWARDS

- Qualified JEE Mains (2016)
- Qualified JAM (2019)
- Qualified GATE (2022)
- Awarded **Junior Research Fellowship (JRF)**, from ANRF (Anusandhan National Research Foundation) under the CRG (Core Research Grant) scheme.

DETAILS OF Ph.D. WORK

Thesis Title:	Projection of Wave Power over the Indian Ocean
Supervisor Name and Affiliation:	Dr. Prashant Kumar, Associate Professor, NIT Delhi, India

Brief Description of Ph.D. Work:	My Ph.D. research, entitled " Projection of Wave Power over Indian Ocean" focuses on the wave power energy research, focusing on the Indian Ocean's potential for renewable energy generation. Using climate models like CMIP5, CMIP6 and COWCLIP, I analyze the impact of oceanic warming and climate drivers on wave power along the Indian coastline, aiming to identify optimal locations for sustainable energy production.
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EMPLOYMENT HISTORY

S. No.	Name of the Post	Institution/University	Period (Year)	Brief description of work
1.	Senior Research Fellow	NIT Delhi	(30-08-2024 to ongoing)	Projection of Wave Power along Indian Ocean
2.	Junior Research Fellow	NIT Delhi	2 Years (30-08-2022 to 30-08-2024)	Projection of Wave Power along Indian Ocean

TECHNICAL SKILLS

- **Software's** - MATLAB, Origin, Latex, MS Word, PowerPoint, Excel, Power BI, Mathematica, Grads, Ferret, CDO, ArcGIS.
- **Languages** - MATLAB, Python, C/C++, Fortran, Python.
- **Soft Skills** - Teamwork, Leadership, Event Management, Written & Oral Communication.

PROFESSIONAL ACTIVITIES

PUBLICATIONS

(a) Journal Articles:

1. Kumar, P., **Yadav, A.**, Sardana, D., & Prasad, R. (2024). Extreme wave height response to climate modes and its association with tropical cyclones over the Indo-Pacific Ocean. ***Ocean Engineering***, 296, 116789. **(Published)**
2. **Yadav, A.**, Kumar, P., Bhaskaran, P. K., Hisaki, Y., & Rajni. (2024). Evaluation of ocean wave power utilizing COWCLIP 2.0 datasets: a CMIP5 model assessment. ***Climate Dynamics***, 62(10), 9447-9468. **(Published)**
3. Kushwaha, P., Pandey, V. K., Kumar, P., Sardana, D., & **Yadav, A.** (2024). Projection of mean and extreme precipitation and air temperature over India: a CMIP6 analysis. ***Journal of Water and Climate Change***, 15(6), 2562-2581. **(Published)**
4. **Yadav, A.**, Kumar, P., Kamranzad, B., & Bhaskaran, P. K. (2025). Assessment of projected changes in mean and extreme wave power across the Indian coastal sectors under high-emission climate scenarios. ***Ocean Modelling***, 102592. **(Published)**.

5. Prasad, R., Kumar, P., **Yadav, A.**, Singh, A., Bhaskaran, P. K., Patra, A., & Rajni (2024). CMIP6 Model Evaluation for Wind Speed responses to IOD during Monsoon Season over the Indian Ocean. *Pure and Applied Geophysics*. (Published)
6. Gulia, S., Kumar, K., Kumar, P., Yadav, A., & Rajni. (2025). Navigating Urban Air Pollution: trends and climate links in Indian megacities. *Environmental Quality Management*, e70210. (Published)
7. **Yadav, A.**, Kumar, P., Rajni (2025). Projections of Wind-Sea and Swell generated WP across the Indian Ocean under Future Emission Scenarios. *Ocean Dynamics*. (Submitted)

(b) Conference Proceedings:

1. **Yadav, A.**, Kumar, P., & Rajni (2023). Extreme Wind-Wave Climate and Tropical Cyclone Activity over the Indo-Pacific Ocean. Samudra Manthan'23, CSIR NIO Goa. October, 2023. (Poster)
2. **Yadav, A.**, Kumar, P., & Rajni (2024). Evaluation of Wave Energy using COWCLIP2.0 over the Indian Ocean. World Ocean Science Conference 2024, IIT Madras. March, 2024. (Poster)
3. **Yadav, A.**, Kumar, P., & Rajni (2024). Projected changes in extreme Wave power generated through swell waves. (IEEE Xplore of ICPEICES – 2024). (Oral: Accepted)

PARTICIPATIONS

1. 3rd prize in **Mathematics Seminar** in **Gargi College (2016)**.
2. Certificate of appreciation in **RSD2022** (Research Scholar Day) **NIT Delhi, 2022**.

REFERENCES

Dr. Prashant Kumar

Associate Professor

Department of Applied Sciences (Mathematics), National Institute of Technology Delhi, India

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Date:

Anshu Yadav

